TP 4

Docker

Objectives:

The goal of this TP is to introduce you to Docker containers and to experiment with it.

We will do the following tasks:

- Run a Docker container.
- Build a Docker container.
- Share a Docker container image.
- Make data created by a container persistent.

Evaluation:

These TPs have two parts that are evaluated:

1. Experimental part: call the Professor to verify what you did when it tells you to do so.
2. Assignment: an assignment needs to be done at the end of the class. Please send it by email to your TP teacher **before midnight the day of the TP**:
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The TP must be done in groups of 2 to 3 students. The groups must stay the same for all the next TPs.

I- Launch the getting-started Docker image:

In this TP, we will run a container image containing a tutorial about Docker, and then do some sections of this tutorial. The goal is to understand this tutorial and be able to summarize the different steps you did in a document.

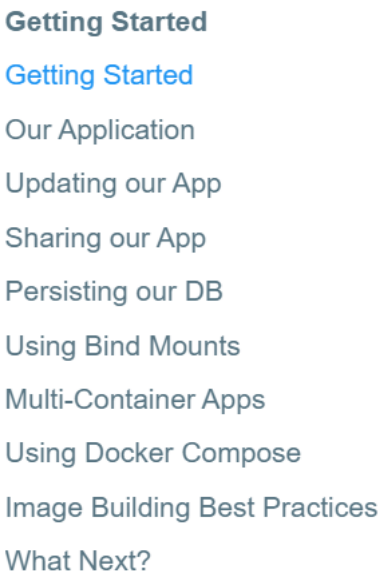
1. First, you have to install the Docker Desktop program from: https://www.docker.com/products/docker-desktop/
2. Then, you have to launch the following command in a terminal to run the getting-started Docker image:

```
docker run -d -p 80:80 docker/getting-started
```

3. Call the professor to show him when the container has run.

II- Build a Docker image:

1. Open the following website: http://localhost/
2. You should now see a tutorial containing several sections:



3. Read quickly the **Getting Started** section (no need to watch the video during the TP, but you can see it at home).
4. Complete the **Our Application** section.
5. Call the professor to show him.

III- Update a Docker image:

1. Complete the **Updating Our App** section.
2. Call the professor to show him.

IV- Share a Docker image:

1. Complete the **Sharing Our App** section.
2. During this section, you will be asked to run your published image in an online Docker playground. Unfortunately, the **Play with Docker** playground no longer works, but you can use the following playground website to complete this part by running your published image, and accessing it from the browser: https://labs.iximiuz.com/playgrounds/docker
3. Call the professor to show him.

V- Make data created by a container persistent:

1. Complete the **Persisting Our DB** section.
2. Call the professor to show him.

VII- Assignment:

Based on what you learned in the TP explain the different steps needed to build a Docker image, run it, share it, and make it persistent. Make sure to explain the different options you use in the commands and use screenshots.